

Proposed Solution Architecture

The proposed solution is an **AI-driven Cybersecurity and Transaction Intelligence Platform** that continuously collects cybersecurity telemetry, banking transaction events, and external threat intelligence from multiple sources. The platform correlates these events using a Security Knowledge Graph and AI-based analytics to identify cyber threats, fraud patterns, and emerging quantum-related risks. It generates context-aware risk scores, provides explainable threat insights, and continuously improves detection accuracy through analyst feedback. The architecture consists of nine major modules, designed to correlate security and transaction events across multiple systems, enabling contextual threat detection instead of isolated alert analysis.

Module 1: Unified Data Ingestion Layer

Objective

Collect cybersecurity telemetry, banking transaction events, and external threat intelligence from multiple sources and convert them into a unified event stream for further analysis.

Responsibilities

- Collect security telemetry from various infrastructure layers.
- Collect financial transaction events from core and perimeter banking systems.
- Support both real-time streaming pipelines and scheduled batch ingestion.
- Validate incoming event structures before processing.
- Maintain data consistency and event chronology.
- Remove duplicate events received from multiple monitoring sources.

Input Sources

Cybersecurity Telemetry

- Firewall Logs
- IDS / IPS Logs
- Web Application Firewall (WAF)
- VPN Logs
- Identity and Access Management (IAM) & Active Directory Authentication Logs
- Endpoint Detection & Response (EDR/XDR)
- Security Information and Event Management (SIEM)
- Email Security Logs
- Data Loss Prevention (DLP)
- Cloud Security Logs

Transactional Behaviour

- Core Banking System
- UPI Transactions
- NEFT / RTGS / IMPS
- ATM Transactions
- Internet Banking & Mobile Banking
- Card Transactions
- Beneficiary Creation / Modification Events

External Intelligence

- Threat Intelligence Feeds
- Malicious IP and Domain Databases
- Indicators of Compromise (IOC) Registries

Working

The ingestion engine continuously receives events from all connected systems through APIs, log collectors, message queues, and real-time streaming pipelines.

Every incoming event undergoes validation to ensure mandatory fields are populated, timestamps are synchronized, duplicate records are stripped, and corrupted frames are discarded before forwarding.

Output

The module produces a **Unified Raw Event Stream**, preserving the original fidelity of the data while enforcing a consistent ingestion envelope for downstream layers.

Module 2: Event Normalization & Threat Enrichment Layer

Objective

To transform heterogeneous data formats and naming conventions into a unified common schema while every event is enriched with relevant threat intelligence and contextual information.

Responsibilities

- Standardize varying log structures into a common schema.
- Normalize and synchronize multi-source timestamps to a baseline standard.
- Correlate user identities across banking systems and security infrastructure.

- Map and associate target infrastructure and devices with specific users.
- Enrich indicators with advanced external threat intelligence.

Working

Each incoming event is parsed into a predefined corporate data model. For instance, source network properties from varying vendors are mapped cleanly:

- **Firewall:** SRC_IP = 192.168.10.5
- **VPN:** Client_IP = 192.168.10.5
- **Internet Banking:** Login_IP = 192.168.10.5

All are normalized instantly to a standard schema attribute: IP_Address = 192.168.10.5.

Threat Enrichment & Context Building

The normalized events are enriched using external threat intelligence to provide additional context before AI analysis.

- **Identify Malware Families:** Tag known signatures to distinct malware classes.
- **Recognize Attack Campaigns:** Cross-reference infrastructure patterns against documented threat-actor profiles.
- **Identify Suspicious Infrastructure:** Identify suspicious infrastructure such as malicious IPs, proxy servers, and command-and-control (C2) servers.
- **Map MITRE ATT&CK Tactics:** Append initial tactic/technique tags directly to normalized metadata.
- **Match Known Fraud Indicators:** Match known fraud indicators using banking fraud intelligence feeds.

Output

The output is a **Context-Enriched Event Repository**, where every event carries uniform schema definitions coupled with asset, identity, geographic, and macro-threat intelligence.

Module 3: Security Context Graph (Knowledge Graph)

Objective

To serve as the central relationship model of the platform by maintaining connections between users, devices, transactions, systems, and security events instead of treating logs as independent database rows.

Responsibilities

- Build persistent and dynamic relationships between distinct physical and digital entities.
- Transition data storage from flat, isolated events into an interconnected graph structure.

- Map deep transaction-to-infrastructure paths to reveal cross-channel fraud and cyber threats.
- Provide structural context directly to downstream AI correlation and analytics engines.

Working

This module processes enriched events and translates them into a unified **Security Knowledge Graph**. It represents relationships between entities as nodes and edges, charting every operational interaction across the banking ecosystem.

Rather than parsing single events, it logically builds out complex, multi-layered identity maps:

$$\text{User} \rightarrow \text{Device} \rightarrow \text{IP} \rightarrow \text{Session} \rightarrow \text{Transaction} \rightarrow \text{Endpoint} \rightarrow \text{Malware} \rightarrow \text{Beneficiary}$$

By maintaining this continuous network layout, the graph enables downstream layers to instantly trace affected systems during an incident or detect multi-stage fraud campaigns traversing entirely different banking channels.

Output

The module maintains a live **Security Knowledge Graph State**, supplying a rich relational topology to all downstream reasoning systems.

Module 4: AI-Driven Correlation & Reasoning Engine

Objective

To analyze the Security Knowledge Graph using advanced temporal and AI-assisted event correlation, moving away from basic event grouping to reconstruct complete attack timelines.

Responsibilities

- Discover hidden attack paths across seemingly unconnected network layers.
- Reconstruct complete attack timelines using strict temporal correlation.
- Correlate related events using AI models to validate unfolding incidents.
- Build complete security incidents instead of isolated alerts.

Working

Traditional SIEM platforms primarily correlate events using predefined rules. This module applies AI-assisted correlation to discover hidden relationships across security and transaction events:

$$\text{AI Correlation Engine} \rightarrow \text{Builds Event Graph} \rightarrow$$

\text{Learns Relationships} \longrightarrow \text{Finds Hidden Attack Paths} \longrightarrow \text{Creates Incident} \end{array}

Temporal Intelligence & Sequence Analysis

The engine evaluates the precise sequence and timing of events, knowing that the exact order of operations dictates the true nature of a threat.

- **9:01** – Login Verified
- **9:02** – Anomalous VPN Tunnel Established
- **9:05** – Database Read Volume Spikes
- **9:08** – High-Value External Transfer Scheduled

The system performs **temporal correlation** across this timeline to explicitly differentiate standard background operations from highly calculated, rapid lateral movements.

AI Correlation Workflow

Instead of relying on simple rule-matching to trigger a risk alert, the engine runs an active, analytical investigation workflow:

\begin{array}{l} \text{Observe Events} \longrightarrow \text{Correlate Related Activities} \longrightarrow \\ \text{Validate Supporting Evidence} \longrightarrow \text{Build Incident Context} \\ \longrightarrow \text{Generate Threat Hypothesis} \longrightarrow \text{Assign Confidence Score} \end{array}

Output

- Chronological Attack Sequences
- Validated Incident Hypotheses
- Contextually Correlated Core Incidents

Predictive Threat Forecasting (Future Enhancement)

- **Objective:** Forecast upcoming tactical steps within an active incident to enable early interception.
- **Mechanism:** Analyze active paths against historically learned attack progressions:
\begin{array}{l} \text{Credential Theft} \longrightarrow \text{Privilege Escalation} \longrightarrow \\ \text{Database Access} \longrightarrow \text{Transaction Infiltration} \end{array}
- **Action:** Generate explicit warnings (e.g., *"High-value transaction attempt predicted within 3 minutes"*) to prompt proactive blocking measures before the final fraudulent activity occurs.

Module 5: Multi-Domain AI Analytics Layer

Objective

To execute concurrent, domain-specific AI assessments on correlated incidents, using an interconnected model where different risk vectors share information dynamically.

Working (Information Sharing)

Rather than operating independently, the analytics engines share risk signals so that suspicious activity detected by one model strengthens the evaluation performed by the others:

$$\text{Behaviour Analytics} \longleftrightarrow \text{Cyber Analytics} \longleftrightarrow \text{Fraud Analytics} \longleftrightarrow \text{Quantum Analytics}$$

Information Sharing Example: If the *Cyber Threat Engine* flags a spike in credential anomalies, it immediately signals the *Fraud Analytics Engine* to enforce stricter anomaly thresholds on transactions. The resulting composite risk score increases dynamically due to the combined evidence of the models.

5A. Behaviour Analytics Engine

- **Responsibility:** Model baseline normal user behaviour patterns for consumers and employees.
- **Working:** Continuously evaluates behavioral footprints (e.g., normal login hours, standard device fingerprints, historical payment locations). It isolates deviations like impossible travel scenarios to yield behavioral risk metrics.
- **Output:** Behavioural Anomaly Indicators and Risk Sub-scores.

5B. Fraud Analytics Engine

- **Responsibility:** Identify financial fraud patterns across payment channels.
- **Working:** Monitors transaction frequency, beneficiary addition histories, and multi-channel activity sequences (e.g., a digital banking transfer initiated immediately following a physical ATM cash withdrawal).
- **Output:** Fraud Pattern Flags and Transaction Risk Sub-scores.

5C. Cyber Threat Analytics Engine

- **Responsibility:** Identify cyber attacks targeting banking infrastructure.
- **Working:** Tracks indicators matching active lateral movement, account takeovers, brute-force distribution, or domain privilege escalation inside banking infrastructure.
- **Output:** Infrastructure Exploit Profiles and Cyber Threat Risk Sub-scores.

5D. Quantum Risk Analytics Engine

- **Responsibility:** Detect patterns pointing toward emerging, long-term cryptographic risks, specifically possible Harvest-Now-Decrypt-Later (HNDL) activity.
- **Working:** Monitors anomalies linked to future decryption preparation, highlighting suspicious large-scale transfers of legacy encrypted data, unusual bulk read requests on encrypted historical archives, or long-lived encrypted outbound data streams.

- **Output:** Quantum Threat Indicators and Exposure Sub-scores.

Module 6: Context-Aware Risk Correlation & Scoring Engine

Objective

To combine all cross-validated AI risk signals with business, transaction, and customer context to determine a single, prioritized confidence-based risk score.

Responsibilities

- Aggregate sub-scores from the specialized behavioral, fraud, cyber, and quantum layers.
- Evaluate environmental and financial context to adjust priority dynamically.
- Suppress false positives by ensuring multi-vector validation before escalating alerts.

Working

The engine processes incoming risk signals through a strict context-aware evaluation. Instead of assessing threat signals purely on isolated anomaly scores, it weights the overall score against critical operational parameters:

Scoring Factor	Evaluation Criteria
Business Criticality & Asset Value	Is the affected endpoint a public terminal or a core ledger system?
Customer Profile Value	Does the targeted account belong to a highly vulnerable or high-net-worth entity?
Data Sensitivity	Is the active data stream exposing public documentation or PII/cryptographic keys?
Transaction Value	What is the exact fiat value or transaction frequency scale of the active transfer request?
Threat Confidence	What is the cumulative confidence level calculated across the prior reasoning layers?

By factoring in these business variables, a low-severity technical anomaly on a highly sensitive asset is prioritized correctly, while an anomaly on a low-value, isolated asset is appropriately managed to prevent alert fatigue.

Output

- Context-Aware Unified Risk Score
- Prioritized Security Incident Classification
- Definitive Confidence Metric

Module 7: Explainable Threat Intelligence Engine

Objective

To convert AI-generated risk assessments into transparent, clear attack narratives, giving SOC analysts full insight into the underlying evidence.

Responsibilities

- Translate AI decisions into clear, human-readable explanations.
- Provide an end-to-end trace of evidence for every high-risk determination.
- Document explicit root-cause hypotheses to help security analysts investigate incidents faster.

Working

Rather than delivering an unhelpful, opaque score such as Risk Score = 98, this module evaluates the hypothesis path, data graph, and cross-validation flags to output clear, structured explanations:

AI Decision Explanation: "The incident risk score is evaluated at 98 based on a high-confidence chain of four correlated anomalies: First, an initial login was verified from a newly registered device profile via an uncharacteristic geographical IP. Second, this session successfully registered a new high-value beneficiary. Third, simultaneous endpoint telemetry detected active malware memory injection on the initiating device. Fourth, a transaction of ₹5,00,000 was executed within 60 seconds of beneficiary confirmation."

Output

- Transparent Root-Cause Analysis (RCA) Reports
- Explainable Threat Narrative Chains
- Human-Readable Incident Timelines

Module 8: Decision, Response & Continuous Learning Engine

Objective

To generate context-aware response recommendations while using a direct analyst feedback loop to continuously retrain and optimize downstream detection models.

Responsibilities

- Formulate context-aware response recommendations for human validation.
- Route high-priority items directly into Security Orchestration, Automation and Response (SOAR) platforms.
- Execute a continuous machine learning feedback loop using direct analyst inputs.

Working

Based on the threat severity and context-aware risk confidence, the engine compiles tailored response options (e.g., isolating an endpoint, freezing a ledger path, or triggering mandatory out-of-band Step-Up MFA).

Continuous Learning & Feedback Loop

To lower false-positive rates over time, this module embeds a continuous learning workflow directly into the SOC analyst's operational routine:

$$\begin{aligned} & \text{\text{SOC Analyst Evaluates Alert}} \rightarrow \text{\text{Applies Verification Tag}} \\ & \rightarrow \begin{cases} \text{\textbf{True Positive}} & \& \text{\text{Reinforces the detection model}} \\ \text{\textbf{False Positive}} & \& \text{\text{Updates the detection model}} \end{cases} \end{aligned}$$

This ensures that whenever an analyst marks an incident as a false positive, the related event patterns and contextual features are processed as a negative training sample. The platform learns from these resolutions, updating the analytics engines to prevent duplicate false alarms under similar operational conditions.

Output

- Recommended Action Plans
- Validated Upstream Orchestration Tasks
- Optimization Vectors for AI Model Retraining

Module 9: Security Operations Dashboard

Objective

To serve as the primary visual interface for security teams, providing complete visibility into the knowledge graph, AI-generated threat assessments, and explainable AI insights.

Responsibilities

- Provide interactive visualizations of the Security Knowledge Graph and attack chains.
- Display synchronized, temporal timelines of multi-channel incidents.
- Present explainable AI narratives and confidence factors clearly.
- Provide a direct interface for the continuous learning feedback loop.

Working

The dashboard unifies the data streams generated across the architecture into a single workspace. Analysts can review cross-validation metrics, trace relationships inside the Security Knowledge Graph, review explainable threat reports, execute recommended response actions, and log feedback into the continuous learning loop through a unified interface.

Output

- Unified Live Threat Ingestion and Operations Console
- Security Knowledge Graph Visualizer
- Explainable AI Insight Workspace
- Continuous Learning Feedback Interface

End-to-End Workflow

1. **Module 1: Unified Data Ingestion** – Collects heterogeneous telemetry and transaction streams.
2. **Module 2: Normalization & Threat Enrichment Layer** – Maps data to a common schema and adds threat context.
3. **Module 3: Security Context Graph** – Maps relationships between identities, devices, and transactions.
4. **Module 4: AI-Driven Correlation & Reasoning Engine** – Reconstructs attack timelines and constructs incidents.
5. **Module 5: Multi-Domain AI Analytics Layer** – Shares risk signals across behavior, cyber, fraud, and quantum domains.
6. **Module 6: Context-Aware Risk Correlation & Scoring Engine** – Calculates prioritized risk scores based on business criticality.
7. **Module 7: Explainable Threat Intelligence Engine** – Documents root-cause insights in human-readable narratives.
8. **Module 8: Decision, Response & Continuous Learning Engine** – Formulates response plans and manages the feedback loop.
9. **Module 9: Security Operations Dashboard** – Provides visual interface and routes